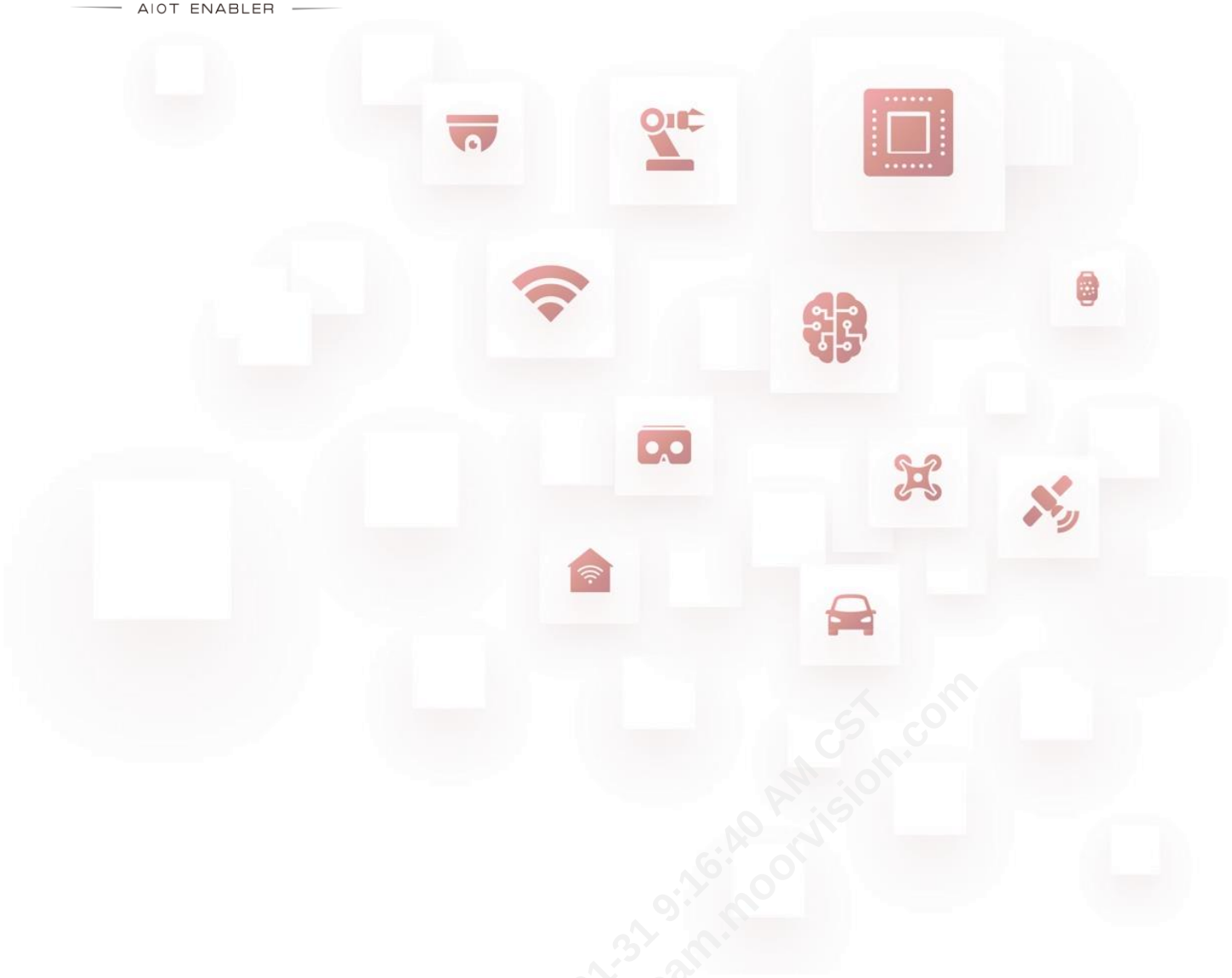
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Thundercomm TurboX C610 Open Kit UVC/UAC Device for LA1.0

User Manual

Rev. V1.0
Sep 12, 2023

DN: tc-g2-231510

Revision History

Revision	Date	Author	Description
1.0	Sep 12, 2023	Chuan Dai	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Overview

This document describes the usage of UVC/UAC device on C610 LA1.0 baseline. The C610 Open Kit board can be used as a USB-UVC camera to transmit data to PC using the UVC/UAC device communication protocol.

➲ **NOTES:**

- As a customized implementation, this solution is not included in the standard C610 LA1.0 SDK.
- In this solution, PC works as the host, while C610 acts as the device.

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Chapter 2. Preconditions

2.1. Prepare test equipment

Step 1. Plug in the IMX415 camera.

Step 2. Plug in a DP cable to the LCD monitor via the DP port.

Step 3. Plug in a USB mouse via the Type-A port.

Step 4. Prepare a Windows PC.

Step 5. Download the *UVC Device Files.zip* file.

Step 6. Unzip the downloaded file. The directory includes:

- *9008driver*
- *qpst.win.2.7_installer*
- *FlatBuild_Turbox-C610_la1.0*
- *UVCCamera.apk*
- *MPC-HC.2.0.0.x64.exe*
- *misc.img*

2.2. Install MPC-HC.2.0.0.x64.exe on Windows

Step 1. Run the *MPC-HC.2.0.0.x64.exe* file.

Step 2. Click the **Next** button.

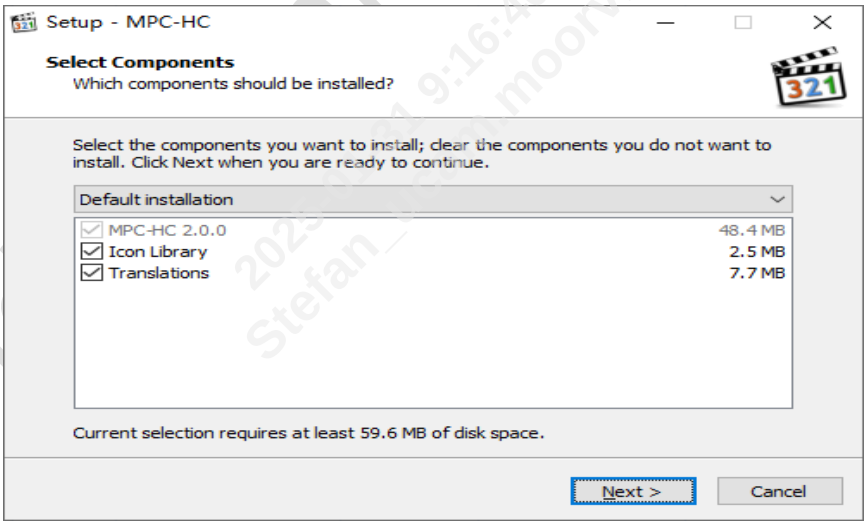


Figure 2-1.Set Up MPC-HC

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Step 3.Click **Next** to continue.

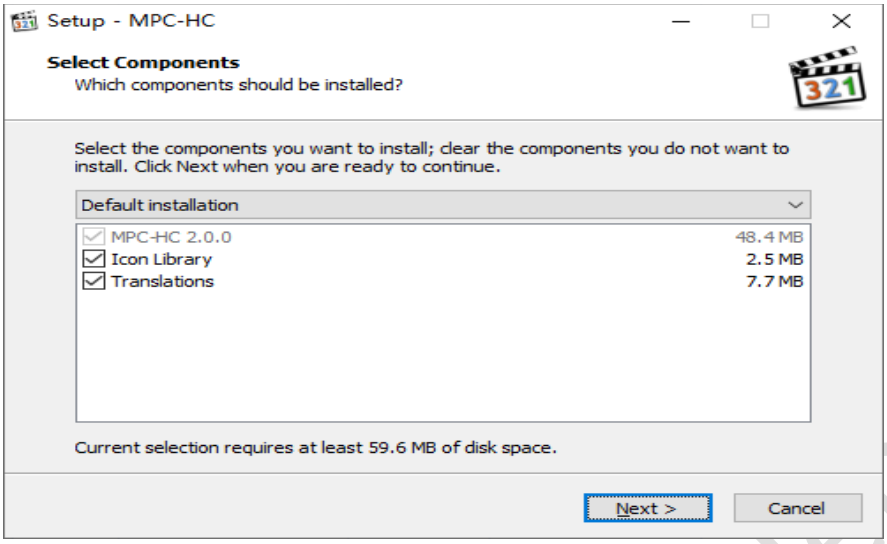


Figure 2-2.Select Components

Step 4.Click the **Install** button.

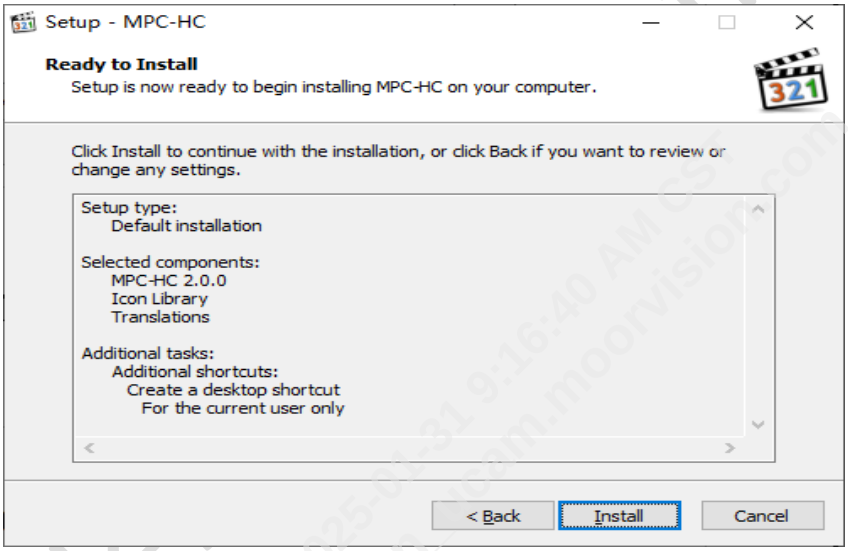


Figure 2-3.Install MPC-HC

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Step 5.Click the **Finish** button.

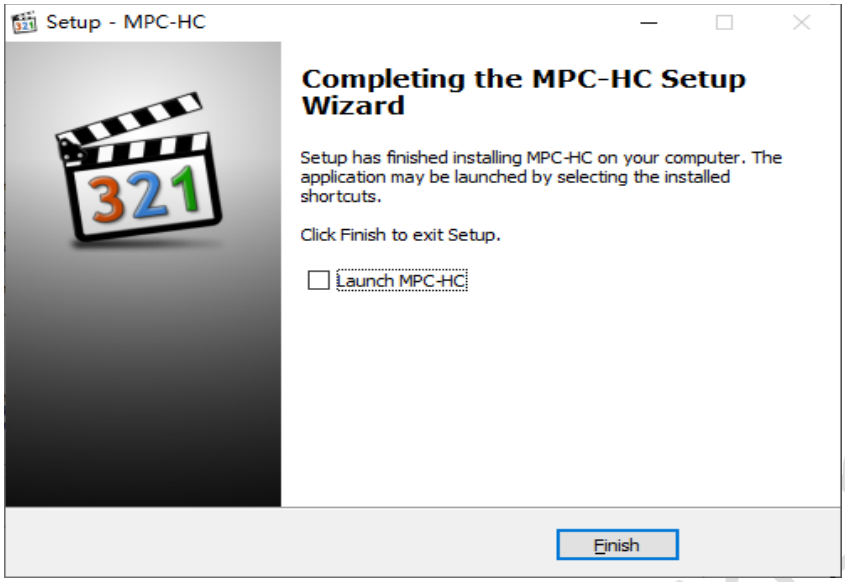


Figure 2-4.Complete MPC-HC Setup

2.3. Install 9008driver on Windows

Step 1.Run the *setup.exe* file, select the checked or free build drivers.

➲ **NOTE:** Checked drivers enable logging.

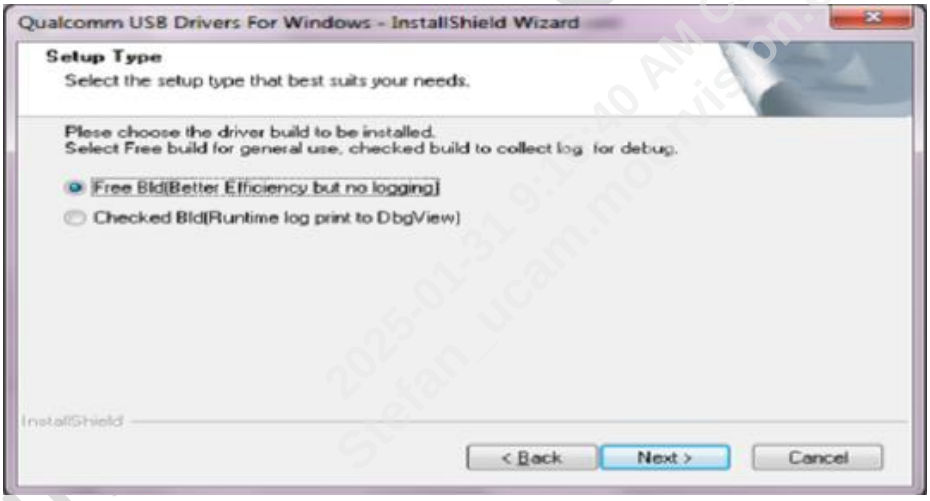


Figure 2-5.Select Setup Type

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Step 2.Click the **Next** button on the welcome screen.



Figure 2-6.Welcome Screen

Step 3.Accept the terms in the license agreement and click **Next**.



Figure 2-7.Accept License Agreement Terms

Step 4.Click the **Install** button.

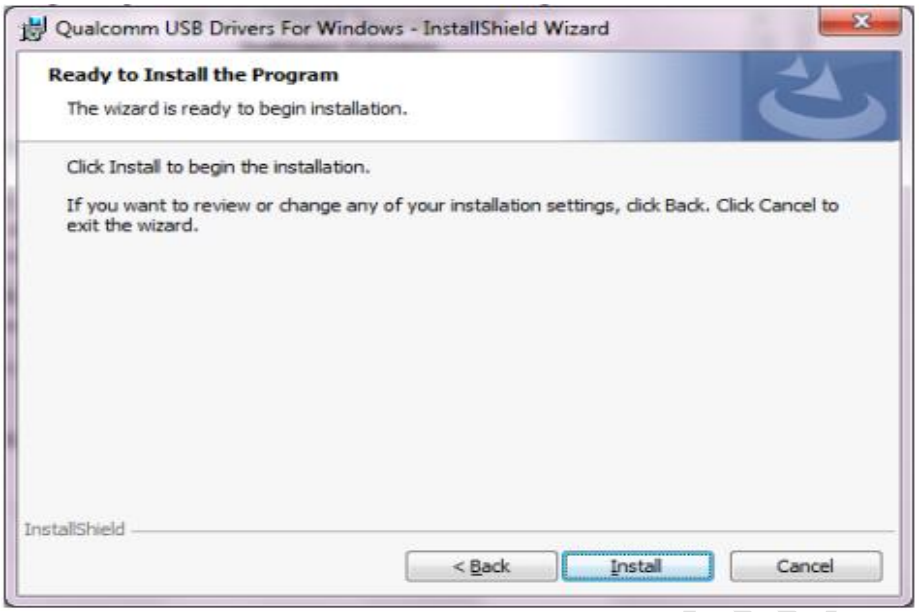


Figure 2-8.Begin Installation

Step 5.Wait for a while, as the installation progress may take several minutes.

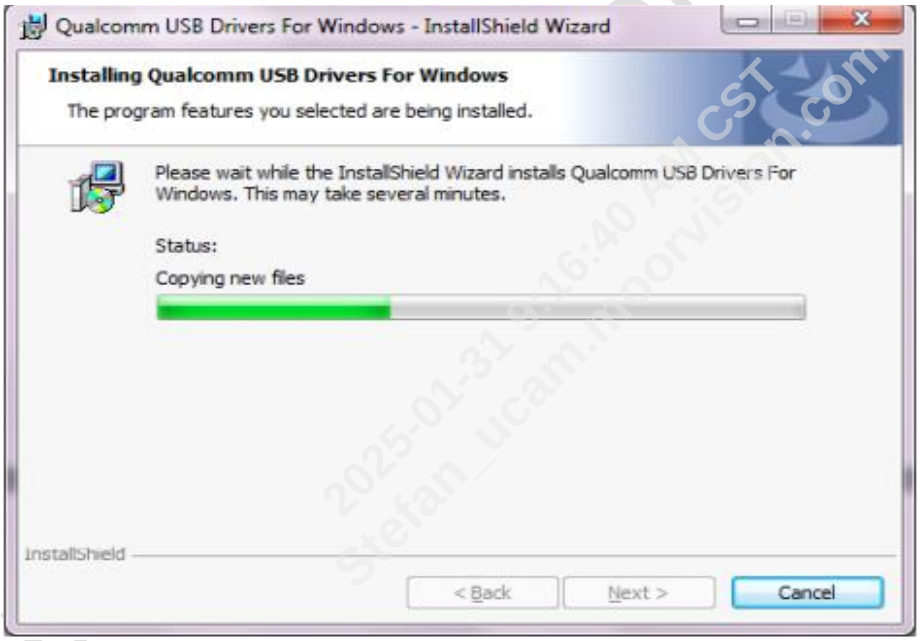


Figure 2-9.Installation Progress

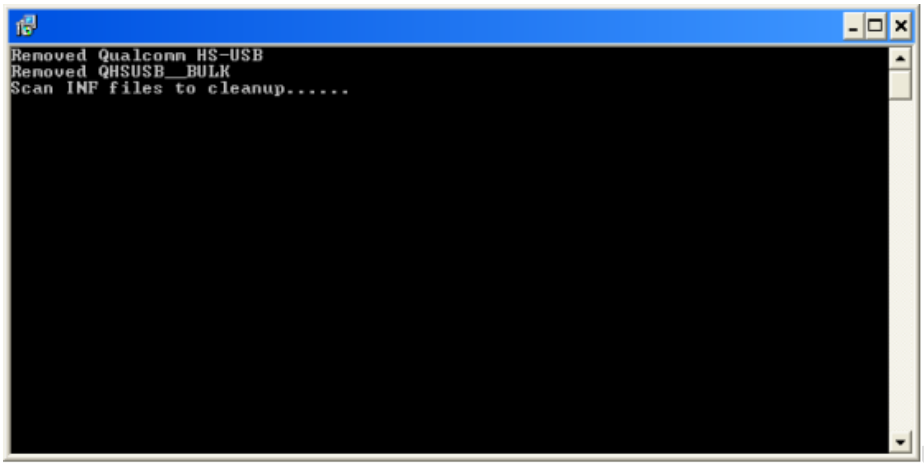


Figure 2-10. Installation Log

Step 6. Upon completion of the installing progress, click the **Finish** button.



Figure 2-11. Completion of Installation

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2.4. Install QPST tools on Windows

Step 1.Run the *QPST.2.7.496.1.exe* file, then click the **Next** button.

Step 2.Accept the terms in the license agreement and continue with the **Next** button.

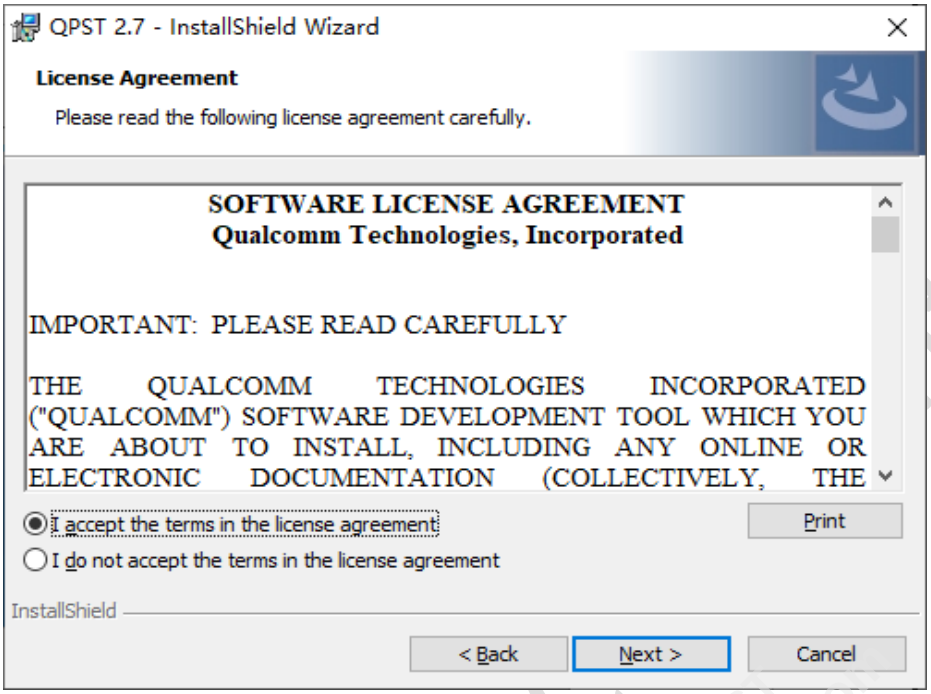


Figure 2-12.Accept License Agreement Terms

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Step 3.Click the **Next** button.

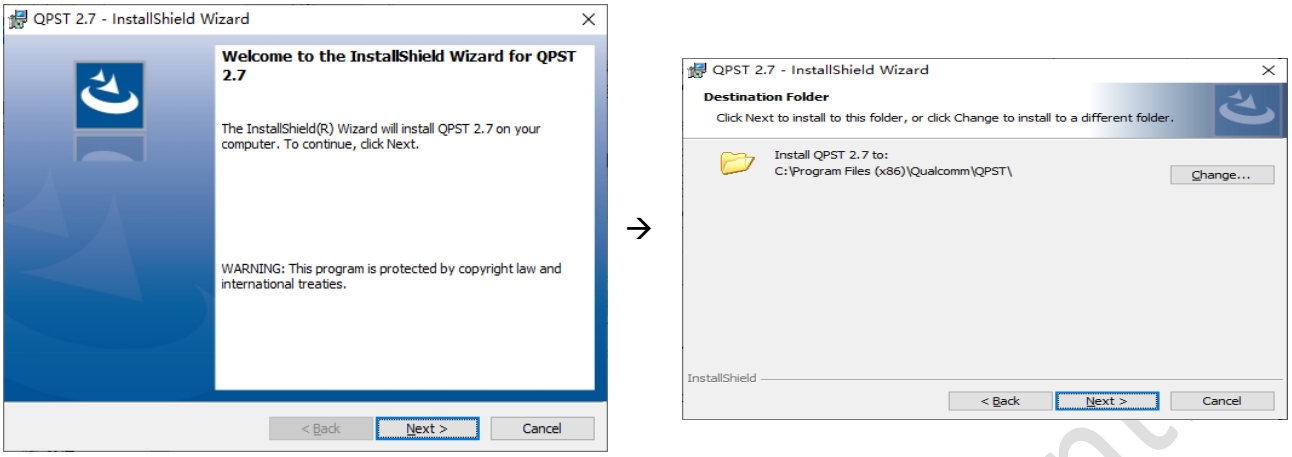


Figure 2-13.QPST Installation Progress

Step 4.Select the "**Complete**" setup type and continue with the **Next** button.

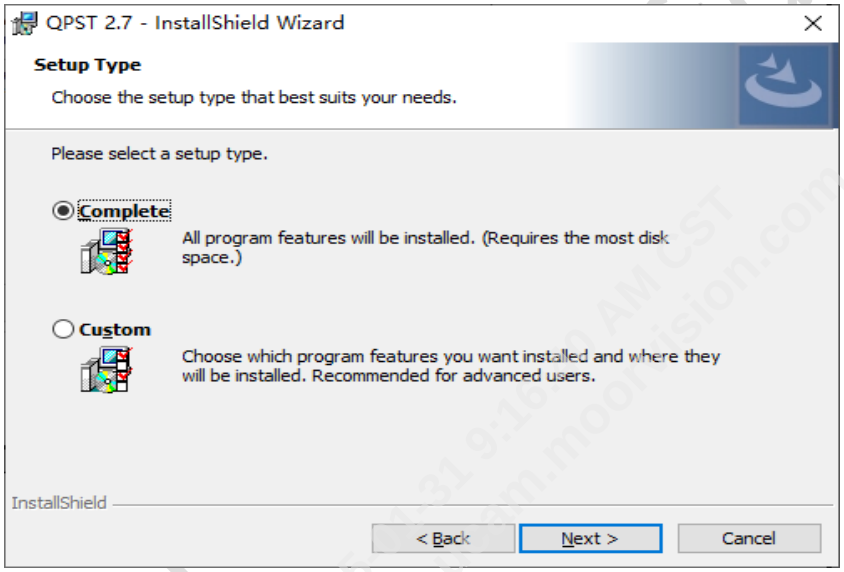


Figure 2-14.Select Setup Type

Step 5.Click the **Install** button to begin installation.

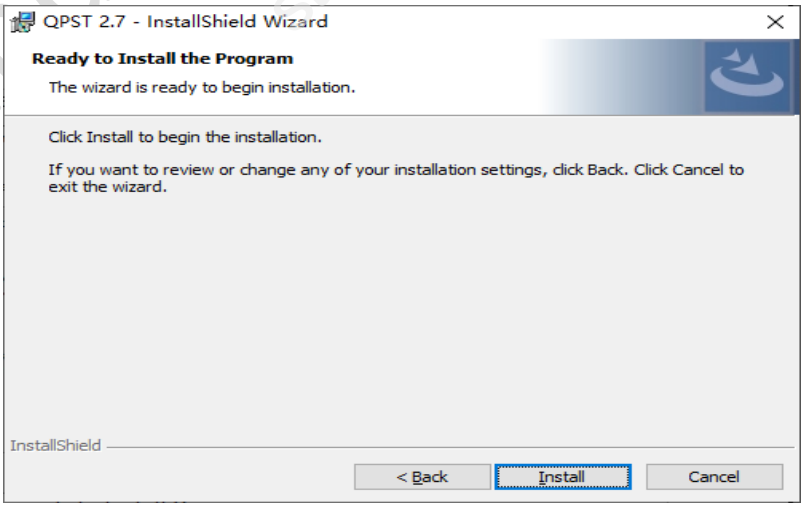


Figure 2-15.Begin Installation

Step 6.Wait a while. Upon completion of the installation progress, click the **Finish** button.

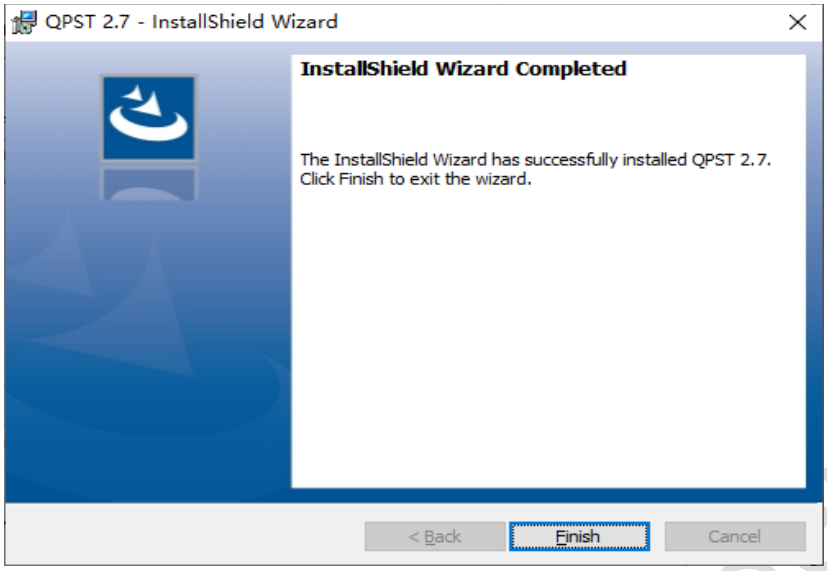


Figure 2-16.Completion of QPST Installation

2.5. Update C610 Open Kit firmware using the QFIL tool

Step 1.Proceed with the sub-steps.

1) Open the **QFIL** software on Windows.
2) Press the **Power on** button of the C610 board.
3) Press and hold the **Force USB Boot** button.
4) Connect the device to the USB port of Windows PC with a USB Type-C cable.
5) Release the **Force USB Boot** button when **QFIL** has detected the com.

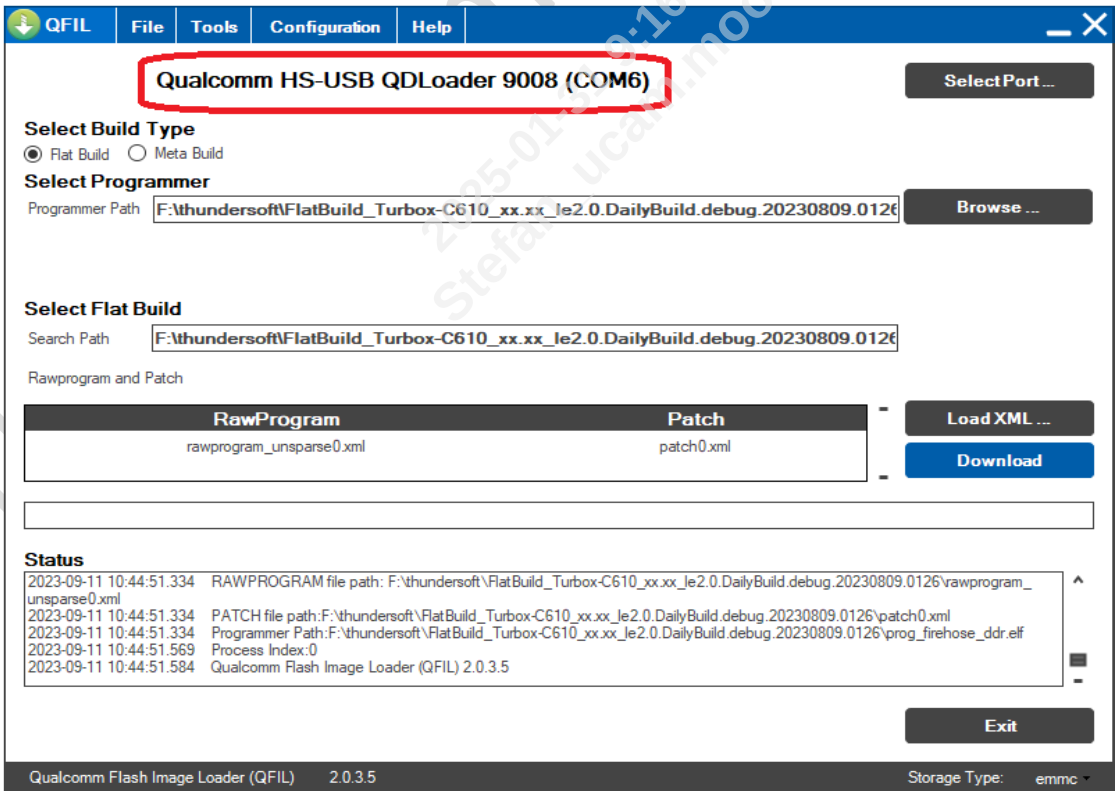


Figure 2-17.Detect Communication Port

Step 2. Make the following selection:

- **Build Type:** Flat Build
- **Storage Type:** emmc

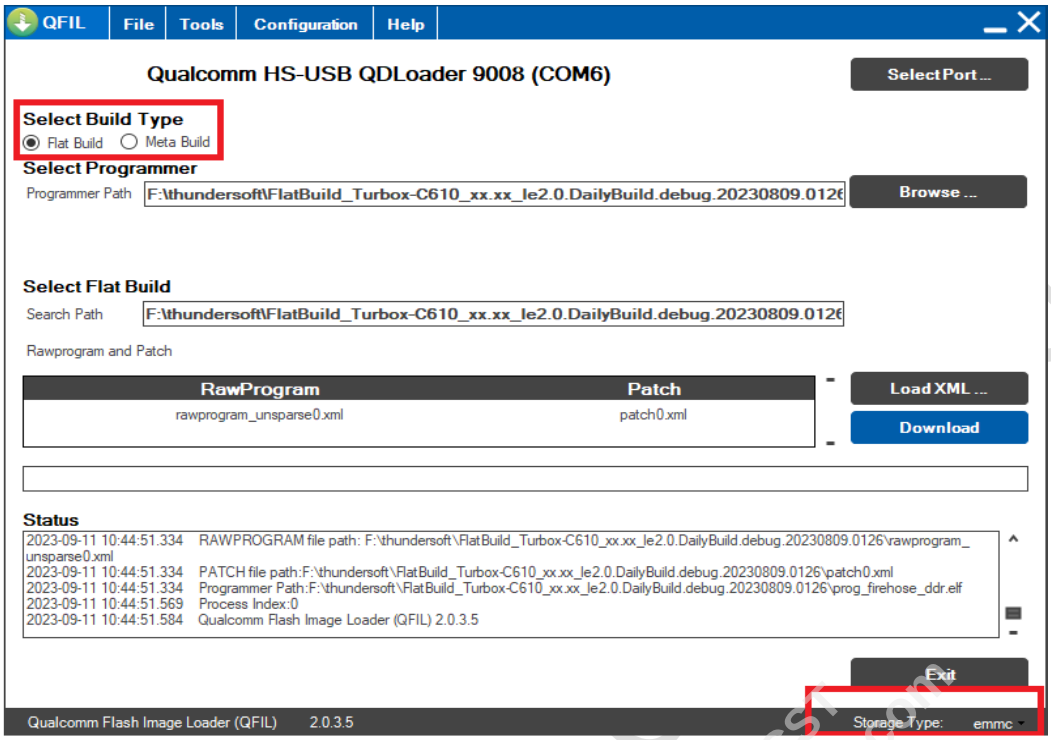


Figure 2-18. Select Build Type and Storage Type

Step 3. Click the **Browse** button and then select the *prog_firehose_ddr.elf* file in the *FlatBuild_Turbox-C610_la1.0* directory.

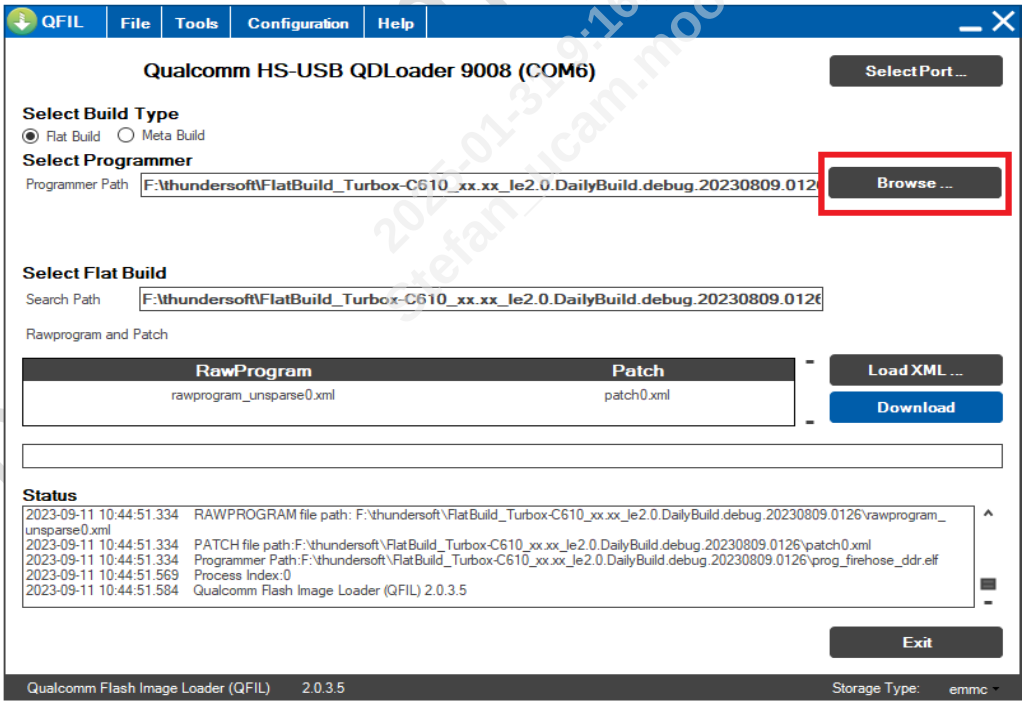


Figure 2-19. Browse File

Step 4. Click the **Load XML...** button and select *rawprogram_unsparse0.xml* and *patch0.xml* in the *FlatBuild_Turbox-C610_la1.0* directory.

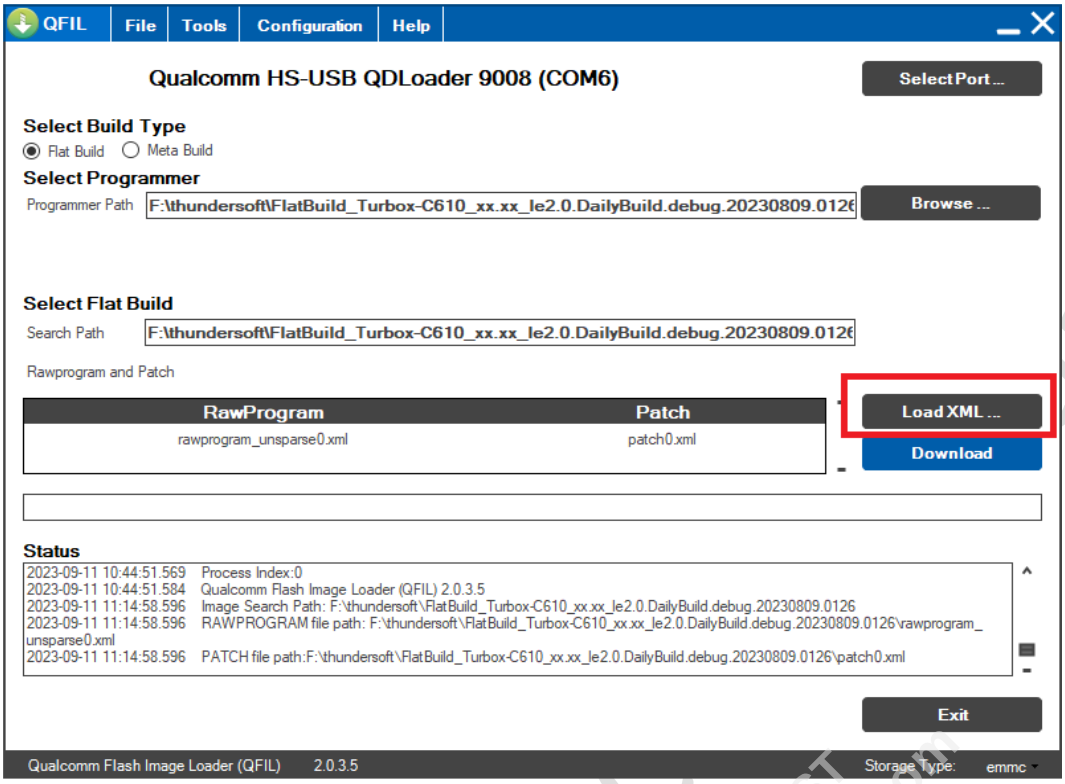


Figure 2-20.Select Files

Step 5. Click the **Download** button to start program update.

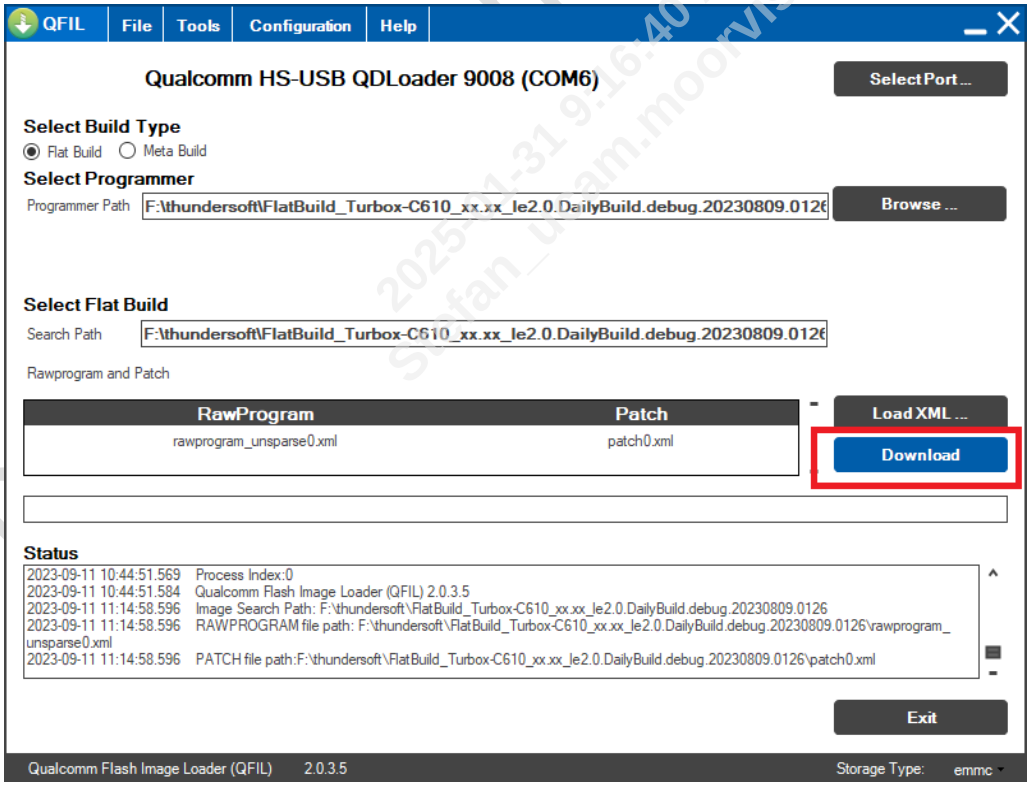


Figure 2-21.Start Download

Step 6. Wait a while. Upon completion of the download progress, power off the C610 board.

2.6. Install UVC Camera.apk on C610

➲ **NOTE:** After the user version is flashed, it is required to switch to **QMMI Mode** to obtain root permission.

Step 1.Allow USB debugging on the DP display as follows:

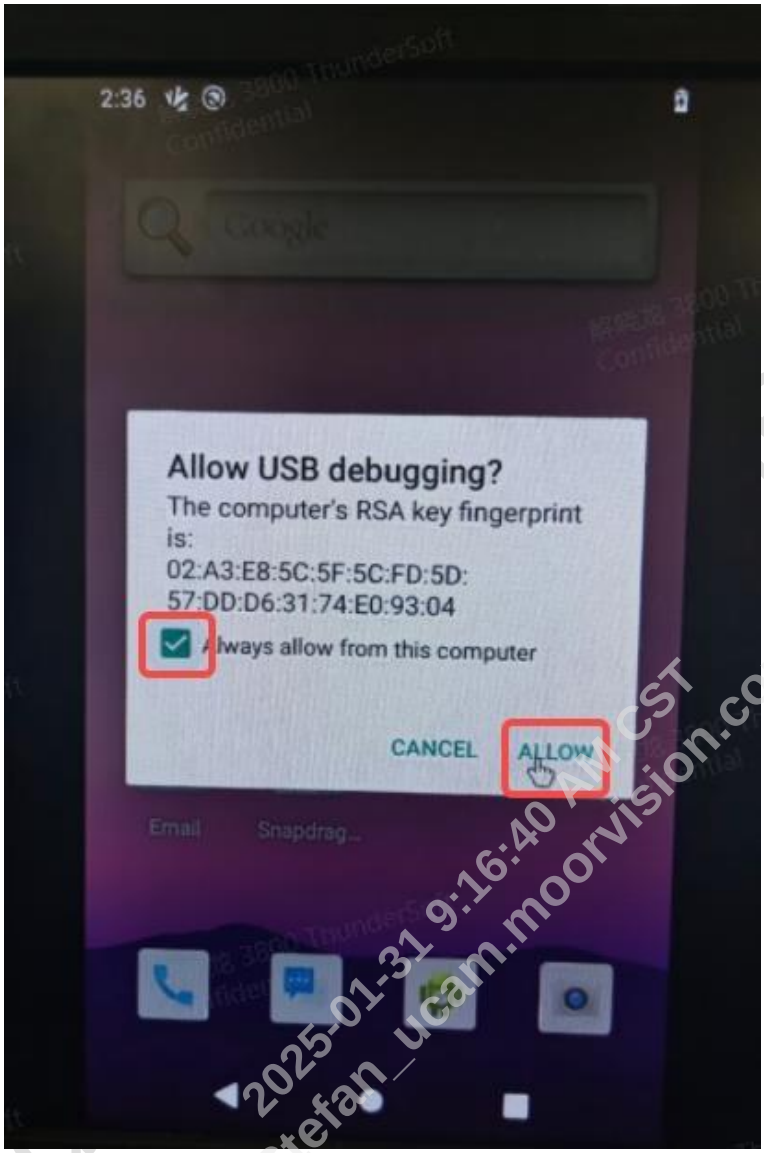


Figure 2-22.Allow USB Debugging

Step 2.Switch to **QMMI Mode**.

```
$ adb reboot bootloader
$ fastboot flash misc misc.img
$ fastboot reboot
```

Step 3.Install UVC Camera apk.

```
$ adb root
$ adb install UVCCamera.apk
```

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Chapter 3. Start UVC and UAC

Step 1. Press the **Power on** button to start the C610 board.

Step 2. Connect the device to the USB port of Windows PC with a USB Type-C cable.

Step 3. Open the **ChiaraCamera** app on the Android screen via the LCD monitor.

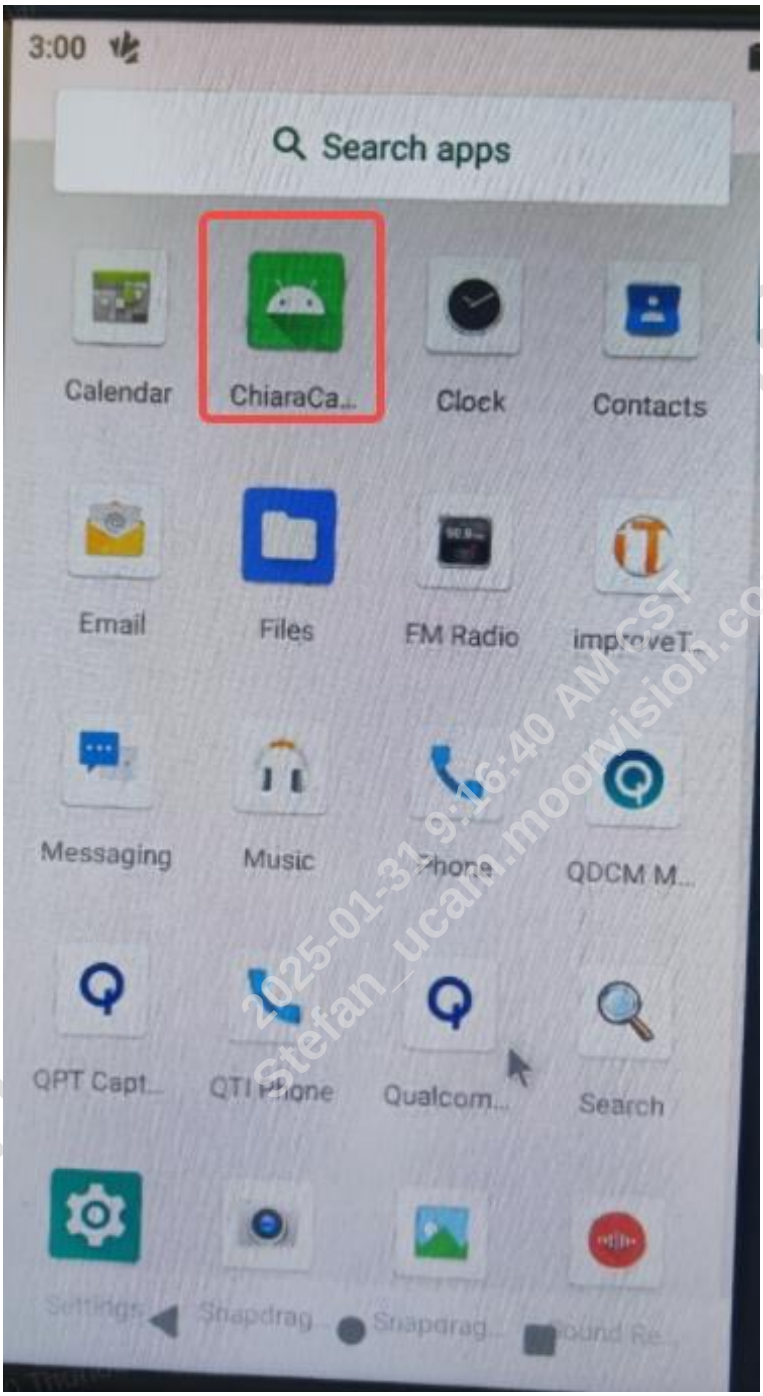


*Figure 3-1. Open **ChiaraCamera** App*

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Step 4.Click the **START** button.

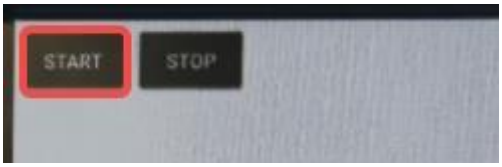


Figure 3-2.Start App

Step 5.Click the **MENU** button.



Figure 3-3.Click Menu

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Step 6.Click the **DOWN** button, and then click the **RIGHT** button to turn on the UVC switch.

➲ **NOTE:** ADB will be disconnected.

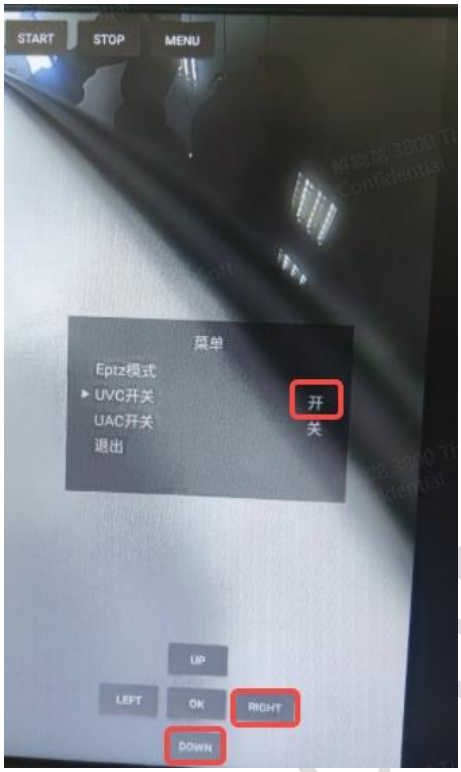


Figure 3-4.Turn on UVC Switch

Step 7.Click the **DOWN** button, and then click the **RIGHT** button to turn on the UAC switch.



Figure 3-5.Turn on UAC Switch

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Chapter 4. Open UVC and UAC Streams

Step 1.Open the **MPC-HC** software on Windows and click **View** -> **Options**.

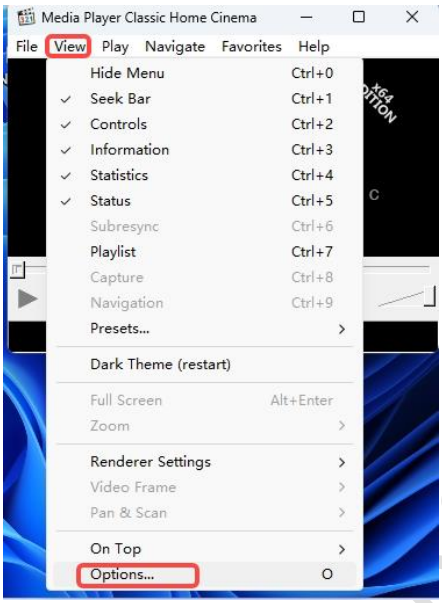


Figure 4-1.Open MPC-HC

Step 2.Click on "**Capture**". Select "**Thudercomm**" as the video and confirm with the confirmation button.

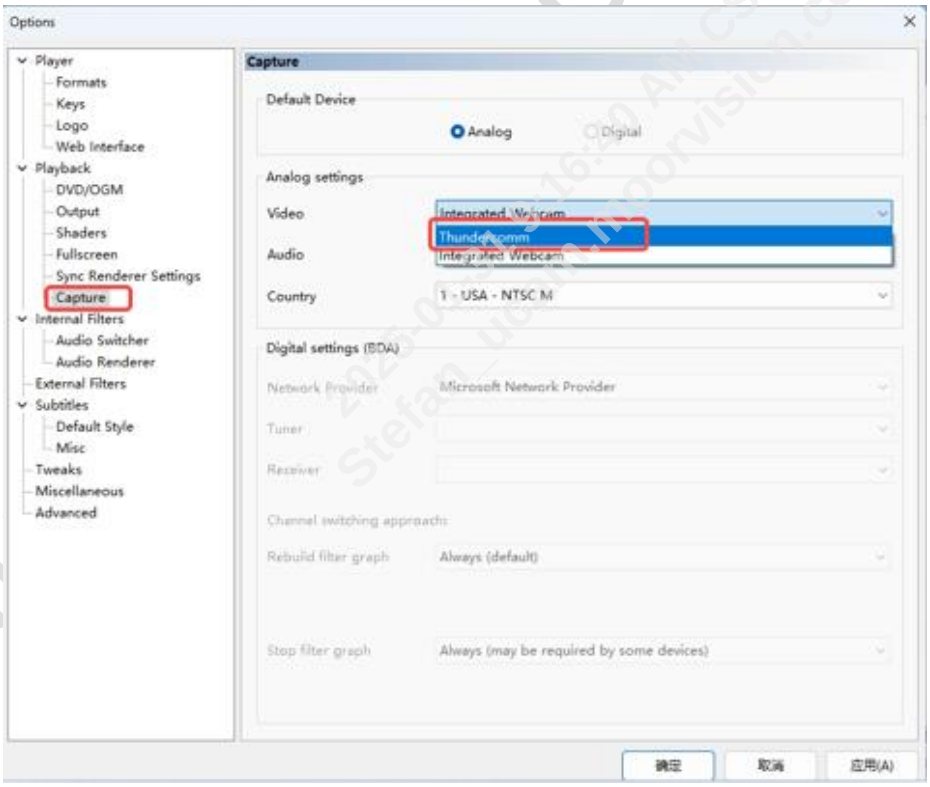


Figure 4-2.Select Video - Thundercomm

Step 3.Click on "**Capture**". Select "Thundercomm (Audio)" as the video and confirm with the confirmation button.

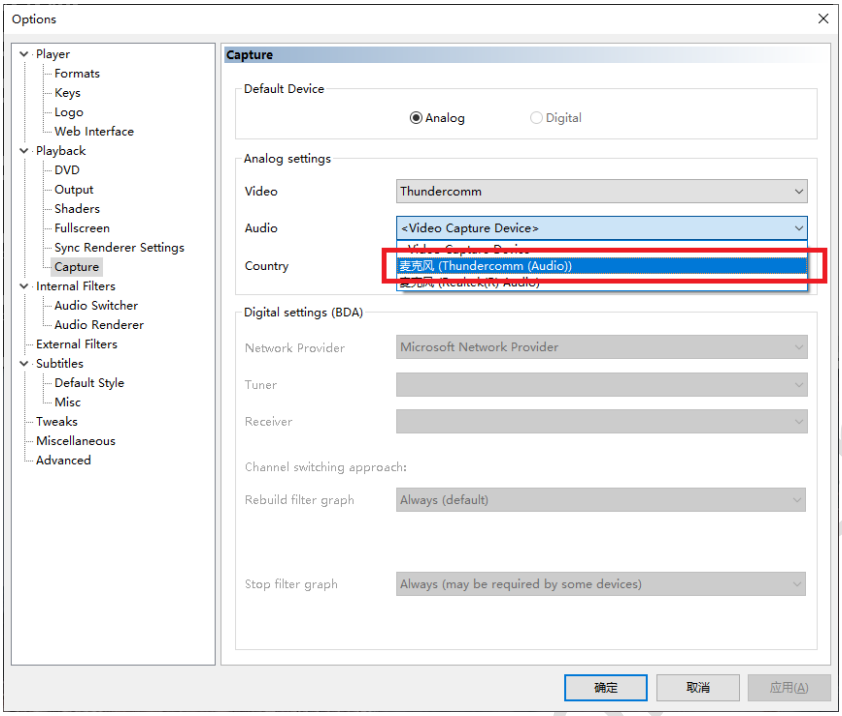


Figure 4-3.Select Video - Thundercomm (Audio)

Step 4.Select **File** -> **Open Device**.

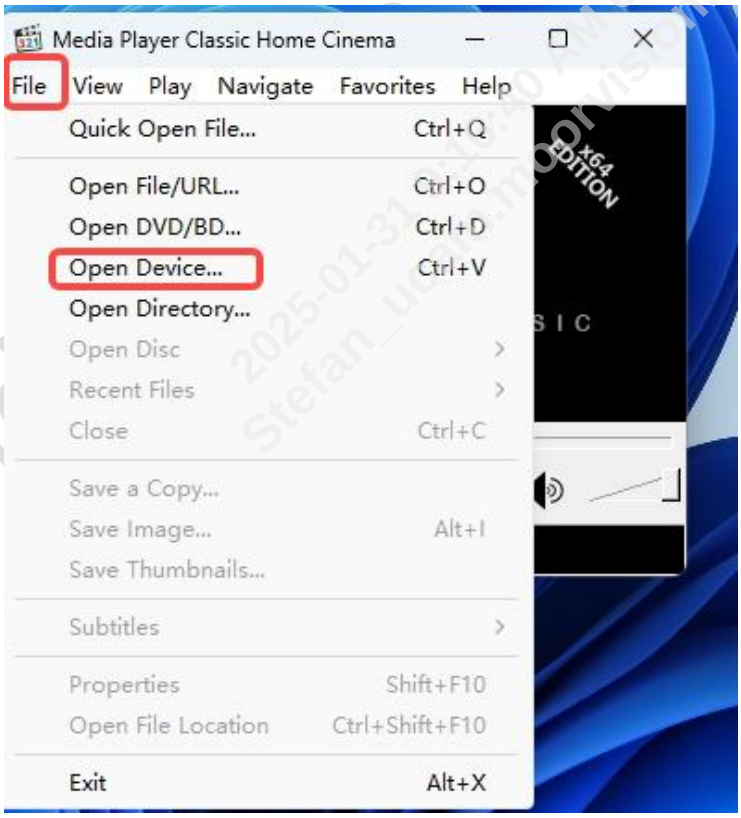


Figure 4-4.Open Device

Step 5.Change the resolution.



Figure 4-5.Change Resolution

Step 6.Change the format.

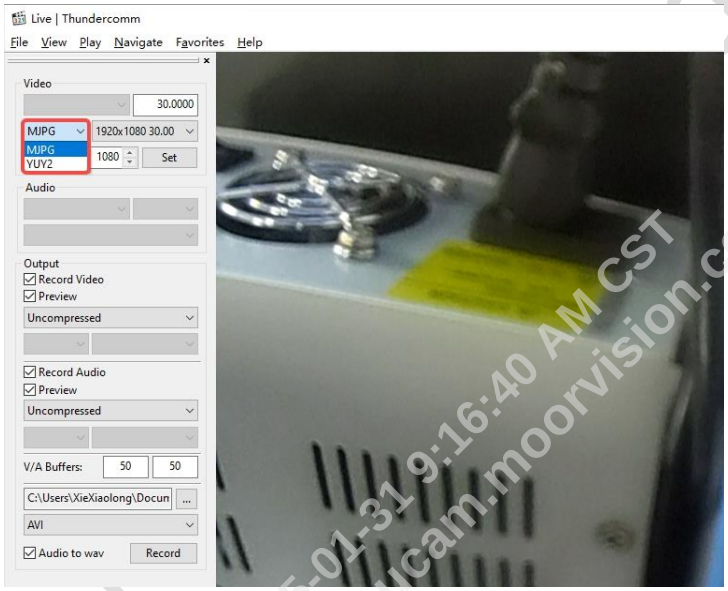


Figure 4-6.Change Format

Step 7.Continue with the [Play], [Pause], and [Stop] operations as required.



Figure 4-7.Test Operations

Chapter 5. Close UVC and UAC Streams

To close the UVC and UAC streams, you can lick the close button on the upper right corner of the MPC-HC screen.



Figure 5-1. Close UVC and UAC Streams

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